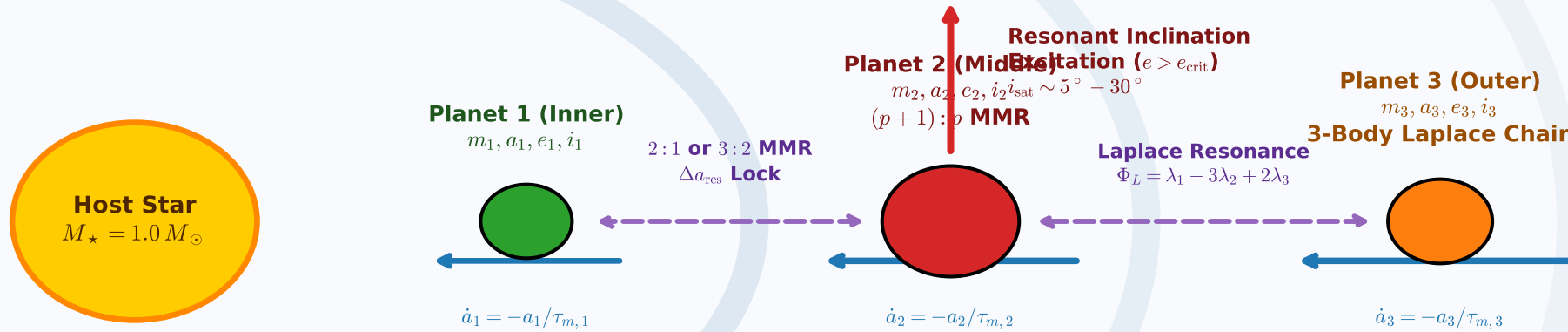


Evolution of Exoplanetary Systems under Gas Drag & Type I Migration

Batygin, Morbidelli, & Tsiganis (2011) — Resonant Chain Locking, Eccentricity Saturation & Inclination Bifurcation



1. Hydrodynamic Dissipative Torques

- $\tau_m = \frac{1}{2C_m} \left(\frac{M_{\star}}{m_p} \right) \left(\frac{M_{\star}}{\Sigma a^2} \right) \left(\frac{h}{r} \right)^2 \Omega^{-1}$
- $\tau_e \approx \left(\frac{h}{r} \right)^2 \frac{\tau_m}{K_{\text{damp}}} \sim 10^{-2} \tau_m$
- $\tau_i \approx 2.3 \tau_e$ (Tanaka & Ward 2004)
- $e_{\text{eq}} \approx \sqrt{\frac{\tau_e}{\tau_m} \frac{1}{(p+1)(1+m_1/m_2)}}$

2. Inclination Bifurcation & Laplace Chains

- $e_{\text{crit}} = \left[\frac{2(h/r)^2}{p+q} \left(\frac{\tau_i}{\tau_e} \right) \frac{|f_{\text{MMR}}|}{|f_{\text{inc}}|} \right]^{1/2} \sim 0.20 - 0.35$
- $\gamma_{\text{inc}} = \frac{3}{4} n_2 \left(\frac{m_1}{M_{\star}} \right) \sqrt{e^2 - e_{\text{crit}}^2} \quad (e > e_{\text{crit}})$
- $i_{\text{sat}} = \arcsin \left(\sqrt{\frac{e^2 - e_{\text{crit}}^2}{1 + e^2 - e_{\text{crit}}^2}} \right) \approx \sqrt{e^2 - e_{\text{crit}}^2}$
- $\Phi_L = p\lambda_1 - (p+q)\lambda_2 + q\lambda_3 \approx 180^\circ$ (Kepler-223 / TRAPPIST-1)